SW Engineering CSC648-848 Fall 2025

Project Tutorall, Milestone 5

Team 04
Matthew Davis (Team Lead) - mdavis09@sfsu.edu
Tan Huynh (Back End Lead)
Anmol Tadikonda (Github Master)
Zainab Abbasi (Front End Lead)
Justine Tenorio

History	
Date Submitted	12/17/2025

1. Product Summary

Product name: Tutorall

Unregistered User:

1. Users shall be able to refine their search by class, professor, category, and price listing.
2. Users shall be able to view a list of tutors and detailed advert prior to registration.
3. Users shall be able to register for services.

Registered Users:

4. Users shall be able to message other registered users directly.
5. Users shall be able to post tutor forms for approval.
6. Users shall be able to remove, and view listings after approval.
7. Users shall be able to see pending tutor listings.
8. Users shall be able to provide documentation for tutor listings.
9. Users shall be able to cancel unapproved listings
10. Users shall be able to view an organized inbox of internal messages.

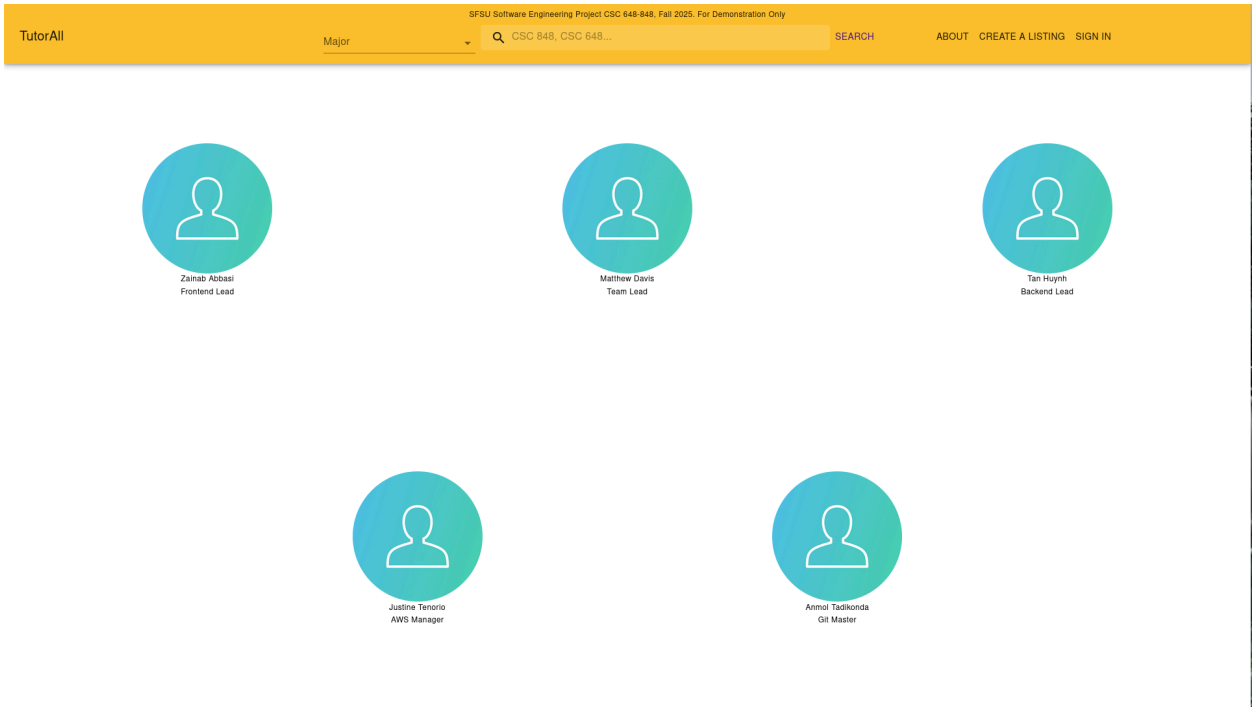
Admin:

11. Admin Users shall be able to approve tutor listings
12. Admin Users shall be able to delete offending content
13. Admin Users shall be able to deactivate users

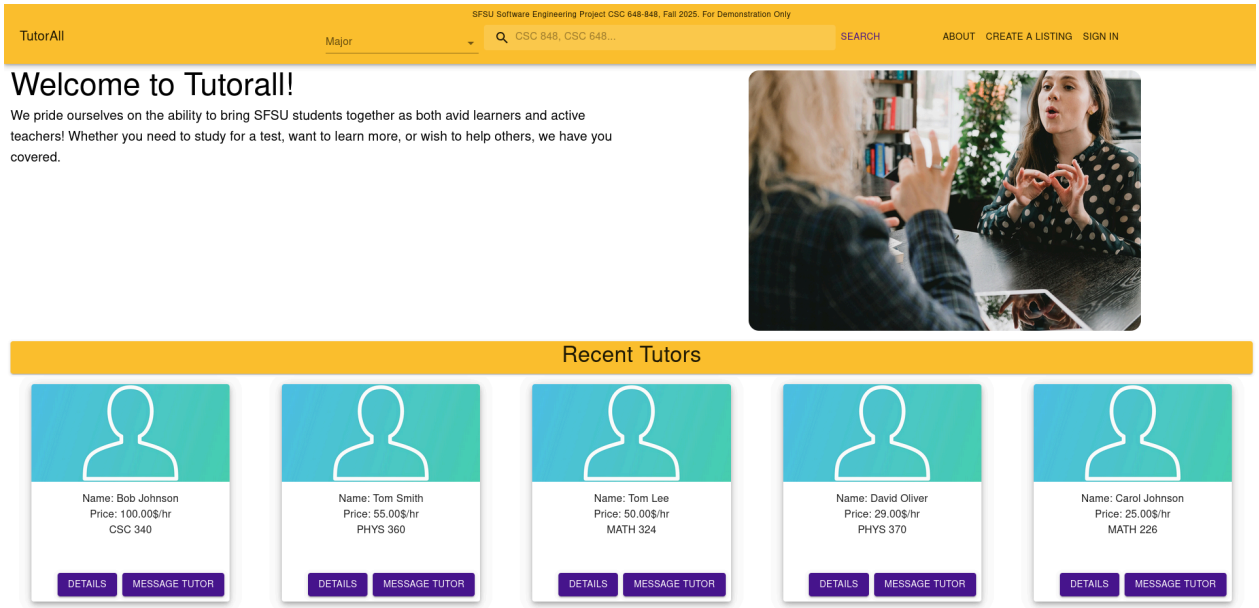
URL: http://http://3.216.62.166

2. Milestone Documents

3. Product Screen Shots



Landing page



Zainab Abbasi

Frontend Lead

Full-stack developer. Frontend lead of the project. Versatile in Computer Science. Pursuing AI.

LinkedIn
GitHub

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TutorAll
Major
CSC 848, CSC 648...
SEARCH
ABOUT CREATE A LISTING SIGN IN

Matthew Davis

Team Lead

A young and scrappy developer who learned a lot from his father about all things electronics. Now a team lead for a group of intelligent and passionent young developers while looking to earn his Master's.

LinkedIn
GitHub

Tan Huynh

Backend Team Lead

Hello, I am the backend team lead.I am currently a senior in university.

GitHub

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SEARCH

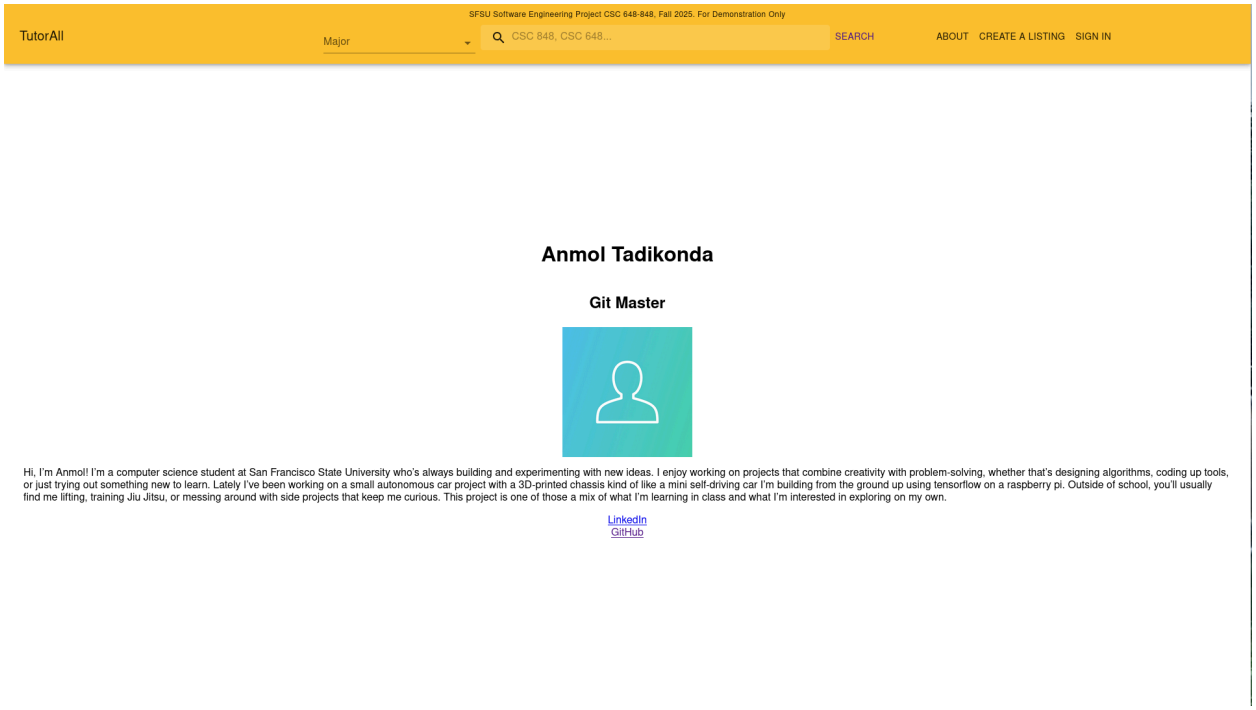
ABOUT CREATE A LISTING SIGN IN

Justine Tenorio

AWS Manager

Senior Comp Sci Student @ SFSU with a strong background in C/C++, Lua, Java, Python and Linux. I enjoy working on hands-on projects with Raspberry Pi, Arduino, and 3D printing. Passionate low-level programming and embedded Linux systems, I'm always looking for more opportunities to expand my skillset and knowledge in order to be a more effective developer.

LinkedIn
GitHub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SEARCH
ABOUT
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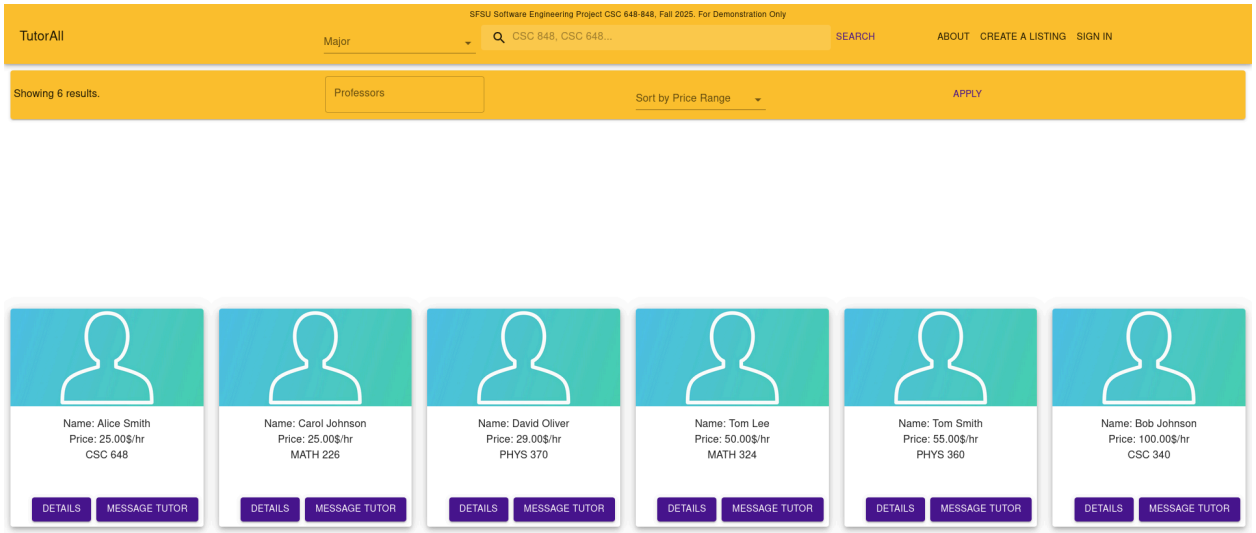
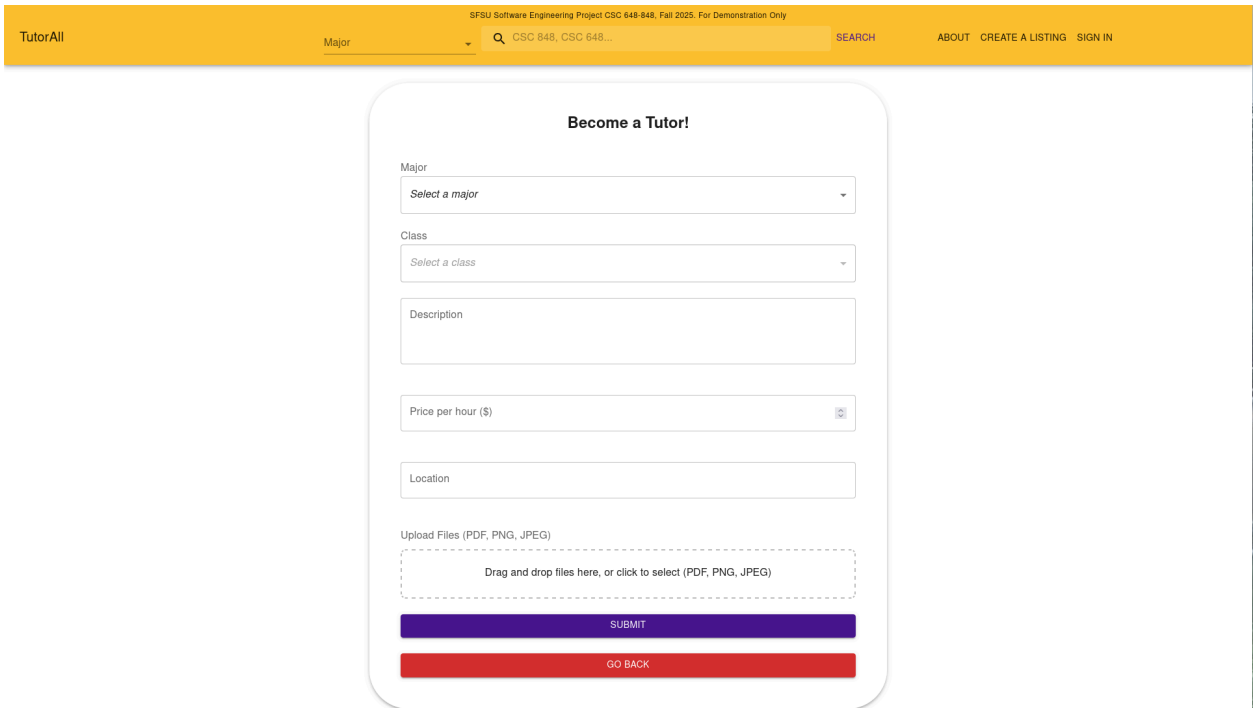
SIGN IN
Anmol Tadikonda
Git Master
Hi, I'm Anmol! I'm a computer science student at San Francisco State University who's always building and experimenting with new ideas. I enjoy working on projects that combine creativity with problem-solving, whether that's designing algorithms, coding up tools, or just trying out something new to learn. Lately I've been working on a small autonomous car project with a 3D-printed chassis kind of like a mini self-driving car I'm building from the ground up using tensorflow on a raspberry pi. Outside of school, you'll usually find me lifting, training Jiu Jitsu, or messing around with side projects that keep me curious. This project is one of those a mix of what I'm learning in class and what I'm interested in exploring on my own.
LinkedIn
GitHub

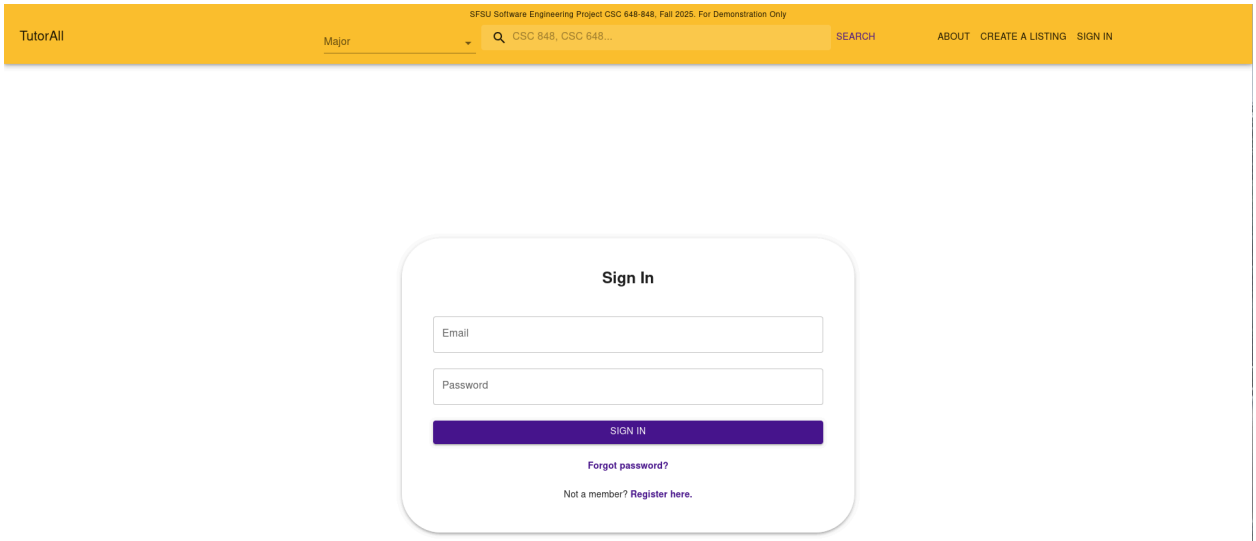
Search


SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only
TutorAll
Major
CSC 848, CSC 648...
SEARCH
ABOUT
CREATE A LISTING
SIGN IN
Showing 6 results.
Professors
Sort by Price Range
APPLY
Name: Alice Smith
Price: 25.00$/hr
CSC 648
DETAILS
MESSAGE TUTOR
Name: Carol Johnson
Price: 25.00$/hr
MATH 226
DETAILS
MESSAGE TUTOR
Name: David Oliver
Price: 29.00$/hr
PHYS 370
DETAILS
MESSAGE TUTOR
Name: Tom Lee
Price: 50.00$/hr
MATH 324
DETAILS
MESSAGE TUTOR
Name: Tom Smith
Price: 55.00$/hr
PHYS 360
DETAILS
MESSAGE TUTOR
Name: Bob Johnson
Price: 100.00$/hr
CSC 340
DETAILS
MESSAGE TUTOR

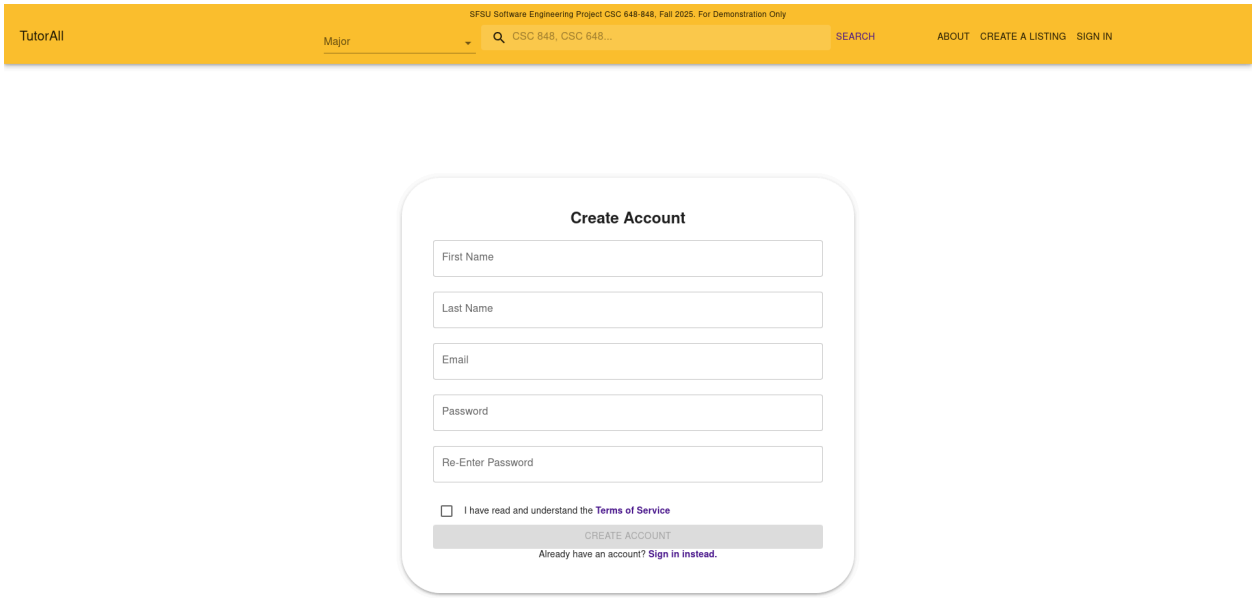
Creating Listing



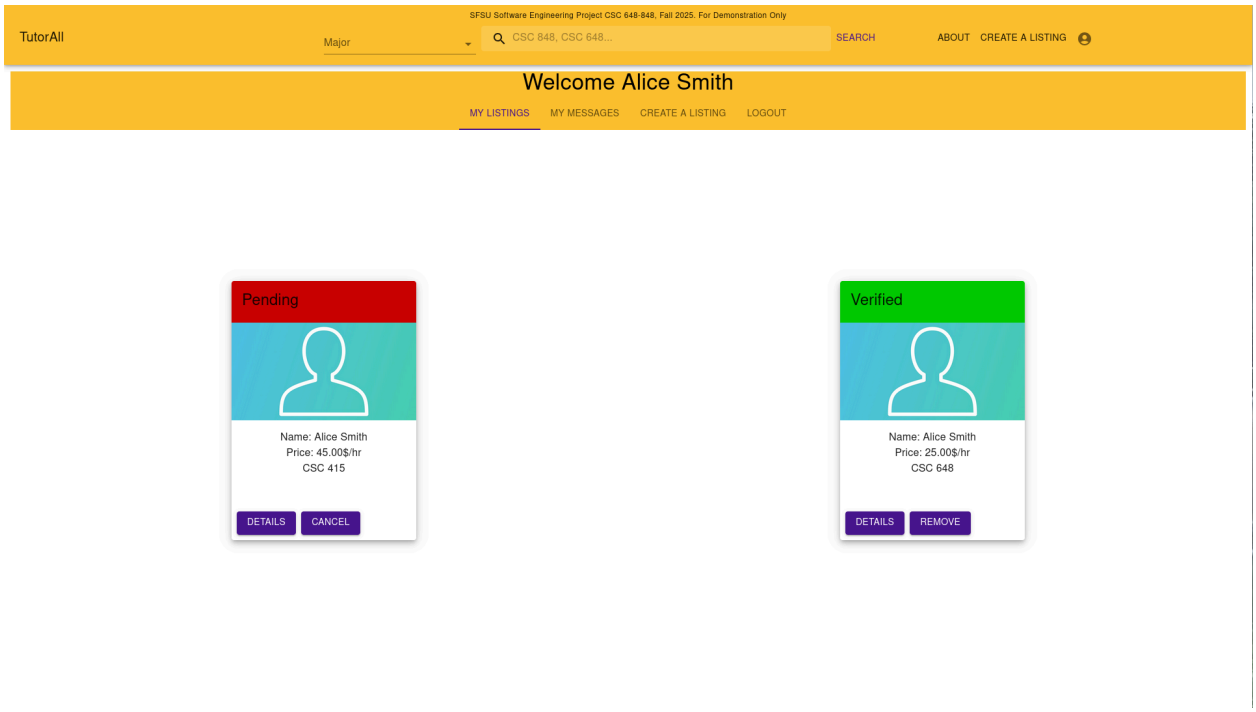
Sign in



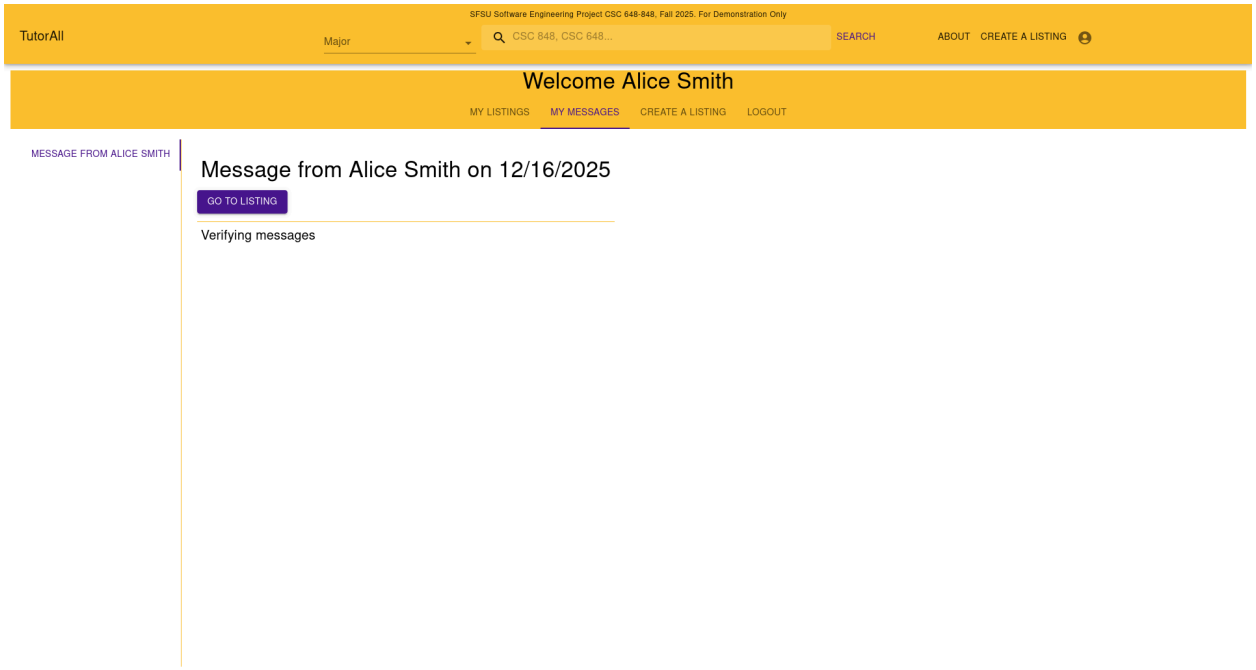
Sign up



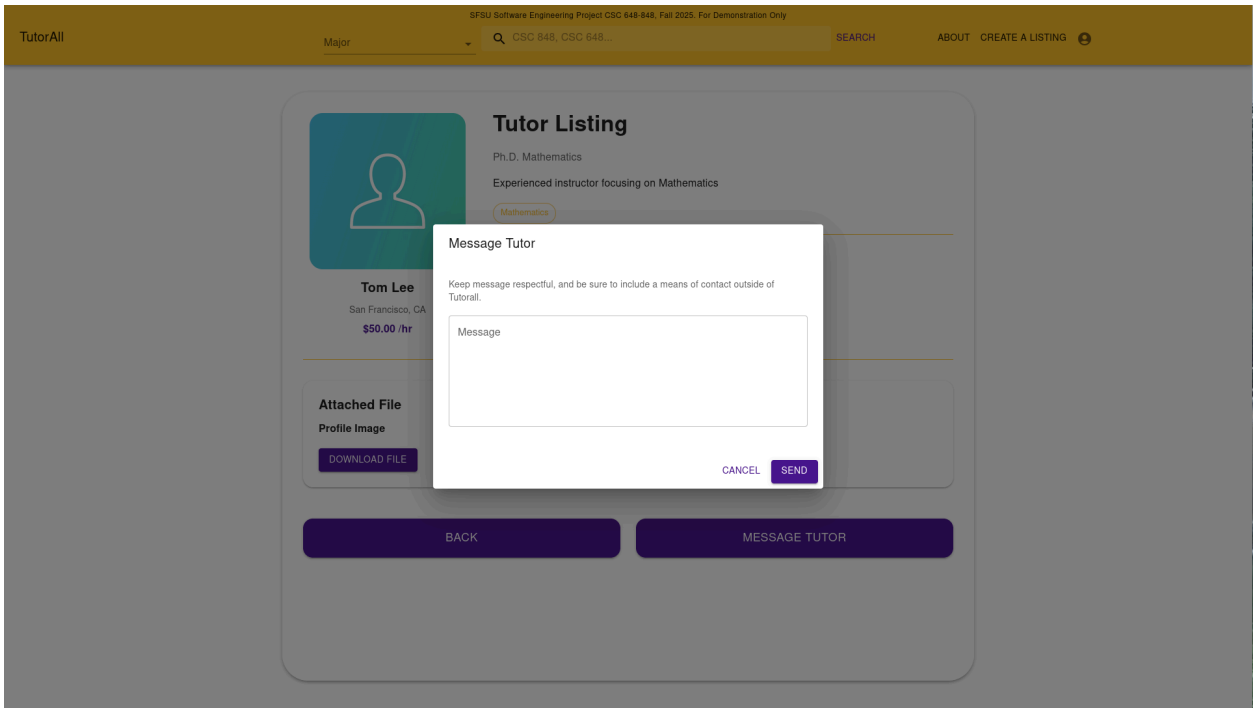
My Listing



My Messages



Messaging a tutor



4. Database Organization

Adverts- Adverts is the tutor listings that are displayed on the website to see if it matches the users needs

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
description	longtext	YES		NULL	
location	varchar(255)	NO		NULL	
price_range	decimal(65,2)	NO		NULL	
verified	tinyint(1)	NO		NULL	
users_id	bigint	NO	MUL	NULL	

Auth Users - DRF users model used to access token authentication protocols

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
password	varchar(128)	NO		NULL	
last_login	datetime(6)	YES		NULL	
is_superuser	tinyint(1)	NO		NULL	
username	varchar(150)	NO	UNI	NULL	
first_name	varchar(150)	NO		NULL	
last_name	varchar(150)	NO		NULL	
email	varchar(254)	NO		NULL	
is_staff	tinyint(1)	NO		NULL	
is_active	tinyint(1)	NO		NULL	
date_joined	datetime(6)	NO		NULL	

Categories- The type of courses that it is under

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
name	varchar(255)	NO		NULL	
courses_id	bigint	NO	MUL	NULL	

Contributors- The information of the team that created the website

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
name	varchar(255)	NO		NULL	
role	varchar(255)	NO		NULL	
image	varchar(100)	NO		NULL	
bio	longtext	NO		NULL	
links	json	NO		NULL	

Courses- The type of classes that are on the website and needed

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
name	varchar(255)	NO		NULL	

Files- The model where the adverts can show off their portfolio alongside with the adverts

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
name	varchar(255)	NO		NULL	
file_type	varchar(100)	NO		NULL	
upload_date	datetime(6)	NO		NULL	
uploader_id	bigint	NO	MUL	NULL	

Professor- Shows all of the teacher that is teaching a particular course

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
name	varchar(255)	YES		NULL	

Users- The users are the ones that registered in the website with a quick description about them

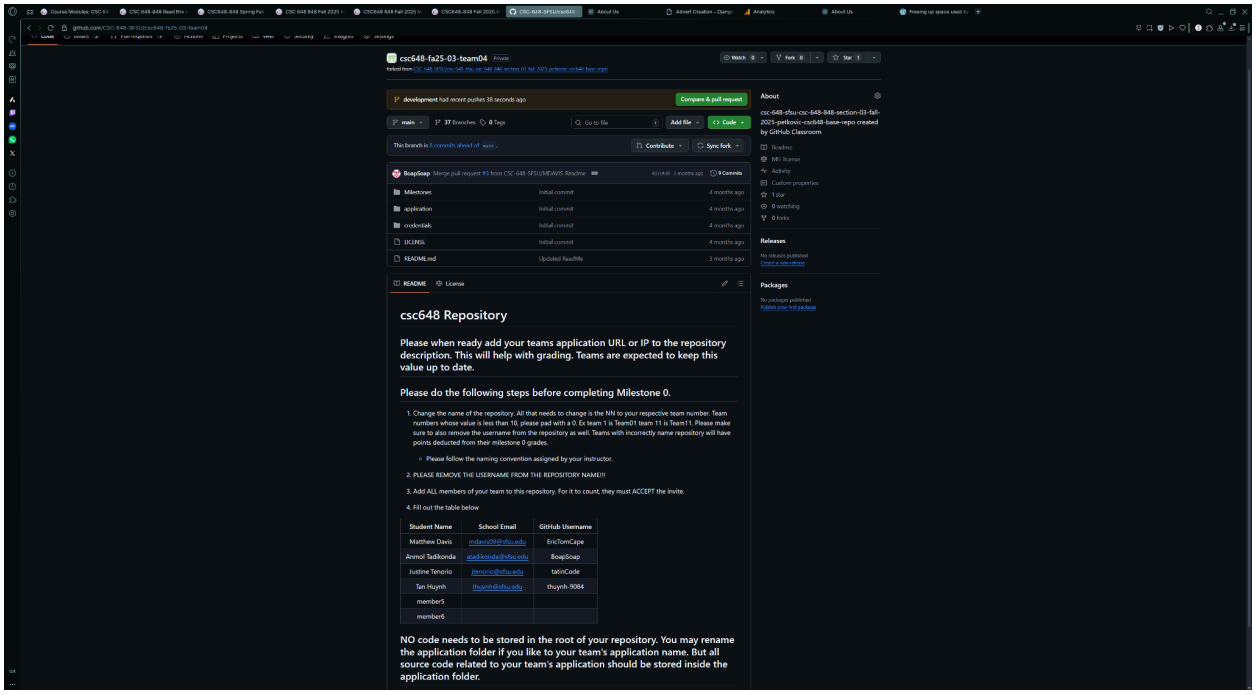
Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
bio	longtext	YES		NULL	
education	varchar(255)	YES		NULL	
location	varchar(255)	YES		NULL	
profile_picture	varchar(100)	YES		NULL	
links	json	YES		NULL	
account_id	int	NO	UNI	NULL	

User_messages- Registered users would send messages to ask for help on a particular subject

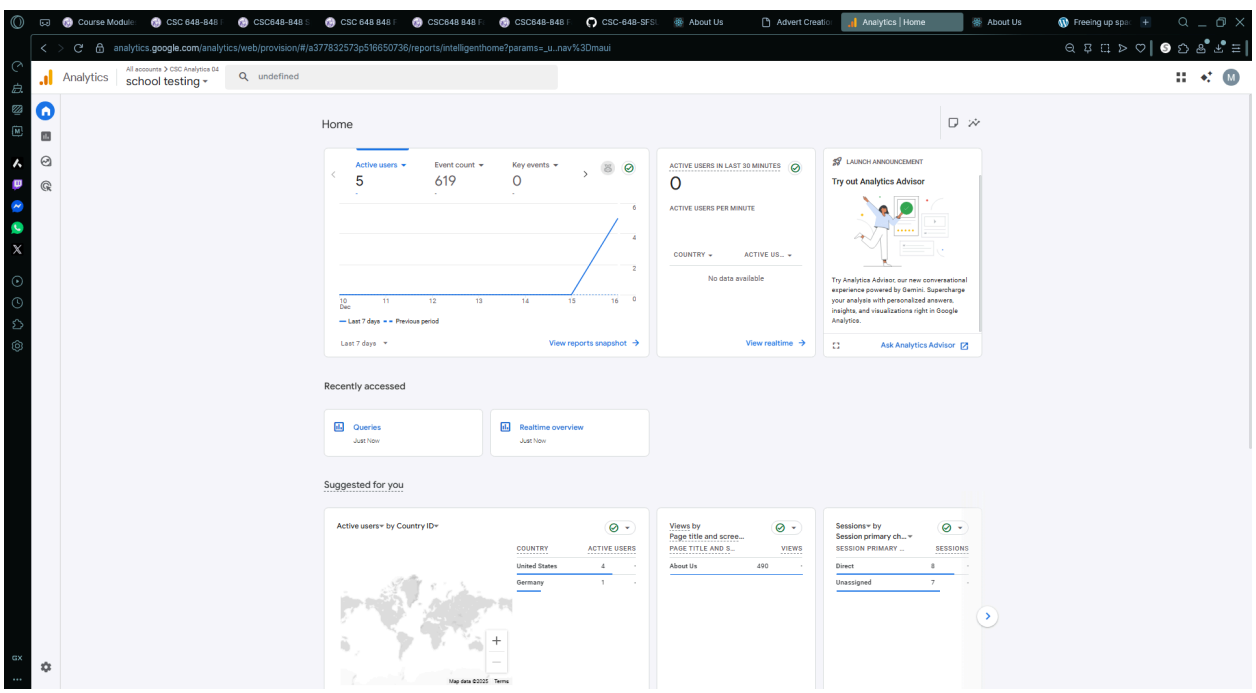
Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
contents	longtext	NO		NULL	
timestamp	datetime(6)	NO		NULL	
read	tinyint(1)	NO		NULL	
advert_id	bigint	NO	MUL	NULL	
receiver_id	bigint	NO	MUL	NULL	
sender_id	bigint	NO	MUL	NULL	

5. Github Organization

a) Main Branches
   i) Main
   ii) Development
   iii) Feature branches
b) Users with Main access
   i) Matthew A. Davis
   ii) Anmol Tadikonda



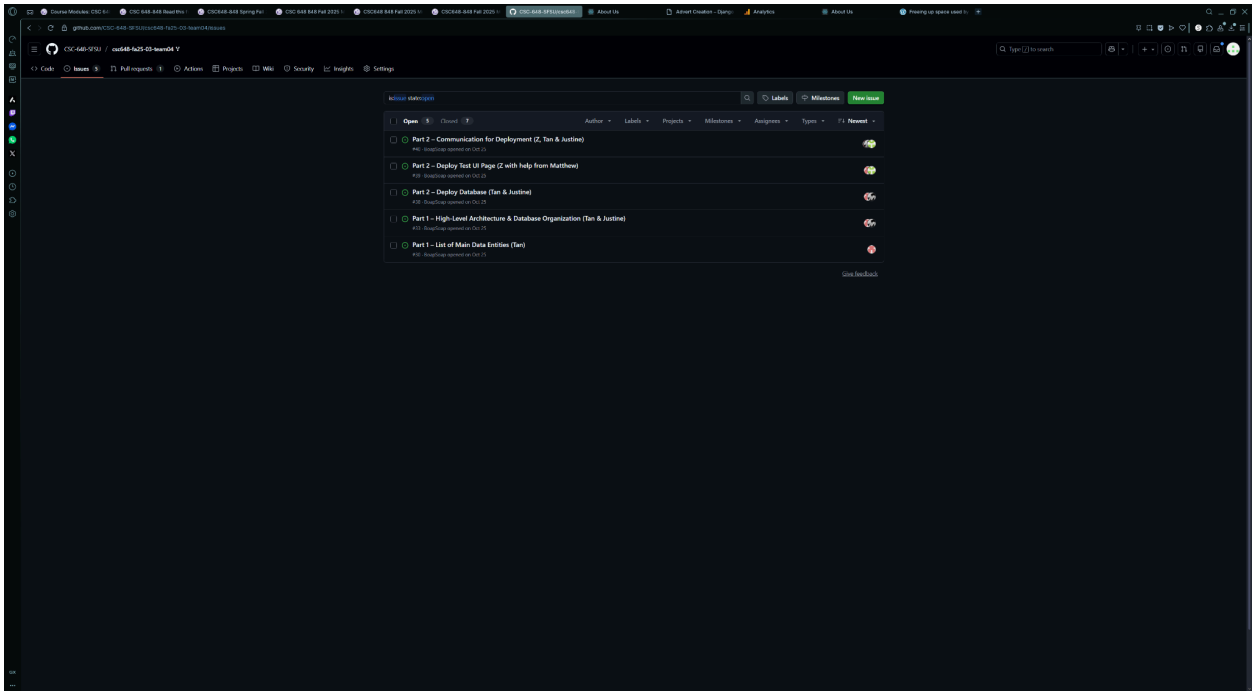
6. Google analytics



7. Project management

We used a mix of Github Issues and Discord in order to manage our requirements and milestones.

Github issues were used to assign actions to individual people and track progress on their work, while we used Discord for ease of communication and as a sort of whiteboard in order to make lists of actions and share intermittent progress reports. One could liken the use to Teams, albeit more informal.



Matthew D. 11/27/2025 9:48 PM

Things To Do

-- General

- ~~Remove gutters @Matthew D.~~
- ~~update app bar @Matthew D.~~
- ~~update advert cards @Matthew D.~~
- ~~fix media @Justine Tenorio~~
- ~~fix DRF @Justine Tenorio~~
- Update adverts model @Tan Huynh

-- ~~Login/Registration~~

- ~~Connect Login form @Matthew D.~~
- ~~Connect registration form @Matthew D.~~
- ~~Add sign out functionality @Matthew D.~~
- ~~Login endpoint @Tan Huynh~~
- ~~Registration endpoint @Tan Huynh~~

-- Advert Details

- Update advert to use live data @Anmol Tadikonda
- Allow for simple file download @Anmol Tadikonda
- Present files as downloadable @Tan Huynh

-- Message Tutor

- ~~Make tutor message a modal @Anmol Tadikonda~~
- Have modal work with live data @Anmol Tadikonda
- Message tutor endpoint @Tan Huynh

-- Advert Creation

- Connect advert creation @Anmol Tadikonda
- ~~Update advert creation to use 1 class @Anmol Tadikonda~~

8. Team member self assessment

Outlook

Re: CSC 648 Self Evaluation

From Tan Huynh <thuynh@sfsu.edu>
Date Mon 12/15/2025 11:32 AM
To Matthew Allan Davis <mdavis9@sfsu.edu>; Anmol Tadikonda <atadikonda@sfsu.edu>; Justine Tenorio <jtenorio@sfsu.edu>; Zainab Abbasi <zabbasi@sfsu.edu>

1. I am the backend lead, so I lead the backend development side of the project with Django. I also implemented the Django models that were first laid out by the team lead. I set up the development and production environment like databases with Docker to have the ease of setting up. Implemented most of the core features that are required for the project. In some of the milestones, I did some documentation like asking if people used GenAI and created a persona of what it feels like to be a tutor using the product.
2. I have about 71 commit submissions to the development branch.
3. Some of the challenges in the team project would be the database migrations provided by Django. This was one of the problems since on one machine it would work but when others are testing and running the program it doesn't work. This caused a lot of headaches since trying to find the exact problem on where it occurred. Also, the second challenge to me is that it is my first time learning Django so it took a lot of time to get used to some of the functions that were already built in the framework, but it was worth learning since it made development faster.
4. My overall experience with GenAI is good since I use it in one of the milestones. It's like generating a picture for me rather than spending a bit more time searching on the web and also finding pictures that do not violate copyright. I thought it was useful, however, throughout the project, I didn't really use it as much. Most of the problems I encountered were just searching for it or asking my teammates to see if they had encountered a similar problem. I mostly see it as an additional assistant in the future and always verify what it says.
5. There are a couple of things that I wish I could have done a bit better, like better time management and organizing my work. There were a lot of times where I could have done some of the assignments much earlier to not stress myself out. For example, I could have spent an hour or two a day during some of my breaks to do some of the smaller assignments. Organization is also important to do next time since it would have helped me break down the tasks to smaller things that are more manageable rather than thinking about the whole picture and implementing. There are some times during development where the code would get really messy and I would get lost in trying to figure out which one works, wasting a bit more time.
6. I have nothing to say.

Outlook

Re: CSC 648 Self Evaluation

From Matthew Allan Davis <mdavis9@sfsu.edu>
Date Mon 12/15/2025 5:15 PM
To Anmol Tadikonda <atadikonda@sfsu.edu>; Justine Tenorio <jtenorio@sfsu.edu>; Tan Huynh <thuynh@sfsu.edu>; Zainab Abbasi <zabbasi@sfsu.edu>

Hello All,

I have to reply to this as well, so.

1) As Team Lead, I have contributed to the front end, back end, deployment, and acted in other leading and supporting roles throughout the project. I created the primary layout, homepage, search page, advert cards, routing, querying setup, context setup, and profile page. I have also contributed in the backend, creating endpoints for profile adverts and profile messages, and advert deletion. I've provided my technical knowledge on how to utilize our given stack, have helped in initial setup, provided tutorials to encourage growth, and researched packages that further our goal. I've also acted as our scrum master, bringing people together to assess our progress, assign tasks, and provide help when asked. I've helped teach best gitflow practices and have helped to review a large portion of the code that has gone into our development branch. I've acted as our primary editor for our Milestone submissions.
2) At the time of writing, I have 108 commits to the primary development branch.
3) There have been two challenges that stick in my mind when it comes to this project. The first is initial setup. I've never deployed to AWS and it's not my particular forte, and so we experience a fair volume of issues trying to set up nginx and routing for our application that should have been resolved sooner. Secondly, I've not had to act as team lead in this kind of setting before, and it's been rather eye opening to the amount of work and effort that goes into coordinating a project of this kind.
4) My overall experience with GenAI is middling. I rarely found a need to use it for any of my assigned tasks, and have a staunch preference against it for image generation. The only places I found it moderately useful was for document summarization, but then I still needed to read through the documentation to make sure it was accurate, so it wasn't much more helpful than reading the document myself to begin with.
5) Overall, I would certainly set much stricter deadlines and more ambitious goals earlier in the project to help prevent any last minute sprints to complete tasks that could have been done much sooner with better forward planning. I would also do a better job communicating exact specifications as soon as possible, which I did try to do but there are many moments where I could have been much clearer on what our goals were sooner.
6) I have nothing more to add here.

Thank you all for your time and effort,
Matthew A. Davis

Outlook

Re: CSC 648 Self Evaluation

From Justine Tenorio <jtenorio@sfsu.edu>
Date Mon 12/15/2025 9:46 PM
To Matthew Allan Davis <mdavis9@sfsu.edu>

Good day, here's my submission for this part of the M5.

1) I am a backend member, and I mostly manage the deployment of our stack. I've also contributed the instructions on how to access the ec2 instance and how to run the server through the docker. More contributions for me would be the high-level architecture as well as listing the roles, created a persona, and I also helped with the data entities and database organization. I also fixed the RAM starvation we were getting early on in the course, in exchange for some storage. I also modified the user messages endpoint to fit our use. And I tried to do my best to keep a good communication between me and my groupmates, I let them know then follow through with it.

2) As of writing this, I currently have 5 commits to the primary development branch. The reason for it being low is because I mainly manage the EC2 instance and the deployment of the development branch. Also most of the fix I make ends up not being pushed by me.

3) Some challenges that we encountered in this project are the constant crashing of the EC2 instance. After a few days of observing and checking logs, it seems to be a RAM issue because we're only running with 1gb of RAM. Another problem is nginx, which could have probably been prevented, have I done my reading of nginx configurations early on instead of after we got the problem. We got 2 major challenge with nginx, one is with the routing of the Django Rest Framework and media files, the other one was more major as it's within the Docker file and nginx config.

4) My experience with GenAI is generally good, for this class my first use of it would be for generating an image for a persona I created. Since then I've been trying to use it as main a search engine, mostly to give me more information for the error I'm getting then checking if it gives a good solution, if not then I look for it myself at least I did not waste too much time figuring out what could be the problem. There's also one time where I got a good answer from it when we were getting errors with our nginx config where nothing was showing, then it gave us a solution which solved the problem, the only problem is it didn't route media properly so no images were showing but it was enough for us to move on to the next feature and we ended up fixing it next time.

5) If I were to do a project, whether that's for the class or personal, I'd try to make a good basis for a plan, could be changed later but at least a layer of the project is done and the next part would be executing it. This removes some of the friction of wondering what to do next, would also be a bonus if I would keep track of a documentation somewhere that way if someone wants to join or contribute, it won't be a frustrating guessing game for them since comments and following the code could sometimes not tell you much. Basically what I took from this class is to be more intentional with what you do, always have a plan, and to list features that are top priority.

Outlook

Re: CSC 648 Self Evaluation

From Zainab Abbasi <zabbasi@sfsu.edu>
Date Mon 12/15/2025 9:49 PM
To Matthew Allan Davis <mdavis9@sfsu.edu>; Anmol Tadikonda <atadikonda@sfsu.edu>; Justine Tenorio <jtenorio@sfsu.edu>; Tan Huynh <thuynh@sfsu.edu>

1) I contributed to the front end, documentation, and the UX design stage. I created 2 UI storyboards, a persona, I coded the early version of some of the website's frontend, and I am currently focused on documentation, so writing summary, test plans, READMEs, etc.
2) 25 commits to the development branch. I was moved to the documentation role later on.
3) I struggled with setting up the project. Historically, I have always struggled with project setup due to many moving parts, errors, etc. Same with here. I also failed to implement a working search function. The best I got was working on my end but failing on other teammates' ends. Issues setting up probably contributed to my failure to code.
4) I used GenAI to help with figuring out UI storyboard panels, finding competitors, and improving test plans. It was useful.
5) I would work with my team to help with setting up.

Outlook

Re: CSC 648 Self Evaluation

From Anmol Tadikonda <atadikonda@sfsu.edu>
Date Mon 12/15/2025 11:55 PM
To Zainab Abbasi <zabbasi@sfsu.edu>; Matthew Allan Davis <mdavis9@sfsu.edu>; Justine Tenorio <jtenorio@sfsu.edu>; Tan Huynh <thuynh@sfsu.edu>

1. I worked primarily on frontend development and served as the Git master for the team. My main technical contributions were focused on frontend features, especially listing creation and related UI functionality. I was responsible for managing the development branch workflow, handling merges, and helping keep the repository organized. Although I was originally assigned to backend tasks, the project needs shifted, and I transitioned into frontend work, which required me to relearn and strengthen my JavaScript skills. I also helped with general debugging and supported teammates when resolving merge or branch-related issues.

2. I made a total of 14 commits to the development branch. While this number is moderate, a significant portion of my time was spent managing Git workflows, coordinating merges, and working on frontend features that required iterative testing and refinement rather than frequent small commits.

3. The biggest challenge for me was getting comfortable with JavaScript again, as I was very rusty coming into the project. Since I was initially expecting to work more on the backend, the shift to frontend development required extra time spent relearning concepts and best practices. Balancing this learning curve with active development and coordination responsibilities made time management more difficult at points in the project.

4. I rarely relied on GenAI during the project. When I did use it, it was mainly for error checking, understanding why certain code was failing, or refreshing my understanding of JavaScript concepts. I did not use it to generate large portions of code, but rather as a support tool to help me learn and debug more efficiently.

5. Next time, I would communicate earlier and more frequently with my teammates about challenges and progress, especially when transitioning roles. I would also try to attend class lectures more consistently. Due to my work schedule, attendance was sometimes difficult, and being in class more often would have made it easier to collaborate with teammates and stay aligned with project expectations.

6. I want to highlight that Matthew was an exceptional team lead. He was very understanding,

patient, and supportive, and he encouraged me to step outside my comfort zone. He gave me the opportunity to push myself in web design and frontend development, which was an area I was not previously confident in, and I am grateful for that experience.